

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
10							***** VALVES *****			817 USE ONLY FOR TIE IN
20							***** MODULAR VALVES *****			B1 ALL BOLTS ARE METRIC DIA.
30										B2 7/8" X 145MM BOLTS ONLY TO BE USED FOR 1
30	U04		MODULAR DBB	K01AC02	0.5	2	MODULAR VALVE DB&B BALL TYPE 150# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
40	U04		MODULAR DBB	M01AC02	0.5	3	MODULAR VALVE DB&B BALL TYPE 150# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO	0	0	FA8 USE THREADED FLANGES FOR INSTRUMENT
50	U04	FS1	MODULAR DBB	M03AC02	0.75	3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	CONNECTIONS ONLY.
60	U04	VAW	MODULAR DBB	M03EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	FS1 USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.
70	U04	VAW	MODULAR DBB	M03EC02	1	1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	NT5 TO BE USED IN NON-PIGGED LINES ONLY
80										VA5 USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
80		VB2					***** BALL VALVES *****			VAW VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
90	U04		BALL	A01AC01	3	4	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO	0	0	VB2 SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
100	U04		BALL	A01AC02	3	4	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO	0	0	VC1 SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
110	U04		BALL	A01AC03	6	6	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB GO	0	0	
120	U04		BALL	A01AC04	6	6	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB GO	0	0	
130	U04		BALL	A01AC07	8	8	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0	
140	U04		BALL	A01AC08	8	8	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0	
150	U04	VA5	BALL	B01AC01	0.75	3	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0	
160	U04	VA5	BALL	B01AC02	0.75	3	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0	
170	U04	VA5	BALL	B01AC03	4	4	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB GO	0	0	
180	U04	VA5	BALL	B01AC04	4	4	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB GO	0	0	
190	U04	VA5	BALL	B01AC07	6	24	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0	
200	U04	VA5	BALL	B01AC08	6	24	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0	
210	U04	VA5	BALL	B03AC01	0.75	2	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0	
220	U04	VA5	BALL	B03AC02	0.75	2	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0	
230	U04	VA5	BALL	A03AC01	3	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO	0	0	
240	U04	VA5	BALL	A03AC02	3	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO	0	0	
250							***** RISING STEM BALL VALVES *****			
250										
260		VA5	BALL	R01AC06	0.75	10	BALL RISING STEM 150# RF LTCS A352 LCC 316SS TRIM METAL RB HO	0	0	
270							***** BUTTERFLY VALVES *****			
270										
280			BUTTERFLY	T01AC61	10	24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0	
290							***** GATE VALVES *****			
290										
300	U04		GATE	G01AC48	0.75	1.5	GATE 150# RF LTCS A350 LF2 CL.1 TRIM 16 WEDGE HW	0	0	
310	U04		GATE	G01AC41	2	8	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW	0	0	
320							***** THROUGH CONDUIT GATE VALVES *****			
320										
330			GATE	W01AC70	3	16	GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0	
340			GATE	W01AC71	18	24	GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0	
350							***** GLOBE VALVES *****			
350										
360	U04		GLOBE	L01AC48	0.75	1.5	GLOBE 150# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW	0	0	
370			GLOBE	L01AC41	2	6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0	
380			GLOBE	L01AC42	8	10	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0	
390							***** CHECK VALVES *****			
390										
400	U04	VC1	CHECK	C01AC50	0.75	1.5	CHECK PISTON 150# RF LTCS A350 LF2 CL.1 TRIM 16	0	0	
410			CHECK	C01AC45	2	6	CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0	
420			CHECK	C01AC52	8	24	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16	0	0	
430							***** PIPE *****			
430										
440										
440			PIPE	5358508	0.75	1.5	PIPE SCH XS SMLS LTCS A333 GR.6 PE	0	0	
450	U04	817	PIPE	5358514	2	2	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0	
460			PIPE	5358515	2	2	PIPE SCH 160 SMLS LTCS A333 GR.6 BE	0	0	
470			PIPE	5358505	3	3	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0	
480										
480			PIPE	9001257	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE	0	0	
490			PIPE	9001254	6	14	LINEPIPE SMLS WT= 6.35MM API-5L X60 BE	0	0	
500			PIPE	9001311	16	18	LINEPIPE SMLS WT=7.92MM API-5L X60 BE	0	0	
510			PIPE	9001485	20	20	LINEPIPE SMLS WT= 8.74 MM API-5L X60 BE	0	0	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE WELDS ABOVE GROUND SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 CODE: ASME B31.8 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 100 °C	12/16/2024	BY OZAC	U04	Page 1 of 4	150K5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
520			PIPE	9001493	24	24	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE	0	0	
530										
530							***** FITTINGS ABOVE GROUND*****			
540										
540			ELBOW	5392600	0.75	2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0	
550			ELBOW	5392438	3	3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6	0	0	
560			ELBOW	9006566	4	4	ELL BW 90 DEG LR WT 6.02 MM A860 WPHY 60	0	0	
570			ELBOW	9002190	6	14	ELL BW 90 DEG LR WT 6.35 MM A860 WPHY 60	0	0	
580			ELBOW	9006478	16	18	ELL BW 90 DEG LR WT 7.92 MM A860 WPHY 60	0	0	
590	U04		ELBOW	9002157	20	20	ELL BW 90 DEG LR WT 8.74MM A860 WPHY 60	0	0	
600	U04		ELBOW	9006471	24	24	ELL BW 90 DEG LR WT 9.53 MM A860 WPHY 60	0	0	
610			ELBOW	9006481	20	20	ELL BW 90 DEG LR WT 8.74 MM A860 WPHY 60-W	0	0	
620			ELBOW	5392606	0.75	2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0	
630			ELBOW	5392444	3	3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6	0	0	
640			ELBOW	9006573	4	4	ELL BW 45 DEG LR WT 6.02MM A860 WPHY 60	0	0	
650			ELBOW	9006571	6	14	ELL BW 45 DEG LR WT 6.35MM A860 WPHY 60	0	0	
660			ELBOW	9006588	16	18	ELL BW 45 DEG LR WT 7.92MM A860 WPHY 60	0	0	
670			ELBOW	9006593	20	20	ELL BW 45 DEG LR WT 8.74MM A860 WPHY 60	0	0	
680			ELBOW	9006583	24	24	ELL BW 45 DEG LR WT 9.53MM A860 WPHY 60	0	0	
690										
690			TEE	5392609	0.75	2	TEE BW SCH XS LTCS A420 WPL6	0	0	
700			TEE	5392447	3	3	TEE BW SCH STD LTCS A420 WPL6	0	0	
710		NT5	TEE	9301610	4	4	TEE BW WT 6.02 MM LTCS A860 WPHY-60	0	0	
720		NT5	TEE	9301609	6	14	TEE BW WT 6.35 MM LTCS A860 WPHY-60	0	0	
730		NT5	TEE	9301606	16	18	TEE BW WT 7.92 MM LTCS A860 WPHY-60	0	0	
740		NT5	TEE	9006487	20	20	TEE BW WT 8.74MM A860 WPHY 60-W	0	0	
750		NT5	TEE	9004210	24	24	TEE BW WT 9.53MM A860 WPHY 60-W	0	0	
760			TEE RED	5392661	0.75	1.5	TEE RED BW SCH XS LTCS A420 WPL6	1	2	
770			TEE RED	5392454	1.5	2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3	3	
780			TEE RED	9006664	2	2	TEE RED BW WT 6.02 MM / SCH XS LTCS A860 WPHY 60	4	4	
790			TEE RED	9006659	3	3	TEE RED BW WT 6.02 MM / STD LTCS A860 WPHY 60	4	4	
800			TEE RED	9006658	3	3	TEE RED BW WT 6.35 MM / STD LTCS A860 WPHY 60	6	6	
810			TEE RED	9006605	4	4	TEE RED BW WT 6.35 MM / 6.02 MM LTCS A860 WPHY 60	6	10	
820			TEE RED	9006014	6	12	TEE RED BW WT 6.35 MM LTCS A860 WPHY 60	8	14	
830			TEE RED	9006017	6	14	TEE RED BW WT 7.92 MM/6.35 MM LTCS A860 WPHY 60	16	18	
840			TEE RED	9006018	16	16	TEE RED BW WT 7.92 MM LTCS A860 WPHY 60	18	18	
850			TEE RED	9004143	8	14	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60-W	20	20	
860	U04		TEE RED	9004123	8	14	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60	20	20	
870	U04		TEE RED	9006100	10	14	TEE RED BW WT 9.53 MM/6.35 MM LTCS A860 WPHY 60	24	24	
880	U04		TEE RED	9006022	16	18	TEE RED BW WT 9.53 MM/7.92 MM LTCS A860 WPHY 60	24	24	
890	U04		TEE RED	9006019	16	18	TEE RED BW WT 8.74 MM/7.92 MM LTCS A860 WPHY 60	20	20	
900			TEE RED	9004149	10	14	TEE RED BW WT 9.53 MM/6.35 MM LTCS A860 WPHY 60-W	24	24	
910			TEE RED	9006483	16	18	TEE RED BW WT 9.53 MM/7.92 MM LTCS A860 WPHY 60-W	24	24	
920	U04		TEE RED	9006484	20	20	TEE RED BW WT 9.53 MM/8.74 MM LTCS A860 WPHY 60	24	24	
930										
930			CAP	5392610	0.75	2	CAP BW SCH XS LTCS A420 WPL6	0	0	
940			CAP	5392448	3	3	CAP BW SCH STD LTCS A420 WPL6	0	0	
950			CAP	9002480	4	4	CAP BW WT 6.02 MM LTCS A860 WPHY-60	0	0	
960			CAP	9005055	6	14	CAP BW WT 6.35MM LTCS A860 WPHY-60	0	0	
970			CAP	9005058	16	18	CAP BW WT 7.92 MM LTCS A860 WPHY-60	0	0	
980			CAP	9005059	20	20	CAP BW WT 8.74 MM LTCS A860 WPHY-60	0	0	
990			CAP	9005060	24	24	CAP BW WT 9.53 MM LTCS A860 WPHY-60	0	0	
1000										
1000			RED CONC	5392458	0.75	1	REDUCER CONC SCH XS LTCS A420 WPL6	1	2	
1010			RED CONC	5392452	1.5	2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3	3	
1020			RED CONC	9006918	1.5	2	REDUCER CONC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4	4	
1030			RED CONC	9006926	3	3	REDUCER CONC WT 6.02 MM / STD WT LTCS A860 WPHY-60	4	4	
1040			RED CONC	9006915	3	3	REDUCER CONC WT 6.35 MM / XS LTCS A860 WPHY-60	6	6	
1050			RED ECC	5392461	0.75	1.5	REDUCER ECC SCH XS LTCS A420 WPL6	1	2	
1060			RED ECC	5392453	1.5	2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3	3	
1070			RED ECC	9001372	1.5	2	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4	4	
1080			RED ECC	9002281	3	3	REDUCER ECC WT 6.02 MM / STD WT LTCS A860 WPHY-60	4	4	
1090	U04		RED ECC	9002510	3	3	REDUCER ECC WT 6.35 MM/STD LTCS A860 WPHY-60	6	6	
1100			RED ECC	9002312	4	4	REDUCER ECC WT 6.35 MM / 6.02 MM LTCS A860 WPHY-60	6	14	
1110			RED ECC	9002298	6	12	REDUCER ECC WT 6.35 MM LTCS A860 WPHY-60	8	14	
1120			RED ECC	9002272	8	14	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	16	16	
1130			RED ECC	9002278	16	16	REDUCER ECC WT 7.92MM LTCS A860 WPHY-60	18	18	
1140			RED ECC	9004131	12	14	REDUCER ECC WT 8.74 MM / 6.35 LTCS A860 WPHY-60-W	20	20	
1150			RED ECC	9004133	16	18	REDUCER ECC WT 8.74 MM / 7.92MM LTCS A860 WPHY-60-W	20	20	
1160	U04		RED ECC	9002292	12	14	REDUCER ECC WT 8.74 MM / 6.35 MM LTCS A860 WPHY-60	20	20	
1170	U04		RED ECC	9002293	16	18	REDUCER ECC WT 8.74 MM / 7.92 LTCS A860 WPHY-60	20	20	
1180	U04		RED ECC	9002251	16	18	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60	24	24	
1190			RED ECC	9004135	16	18	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60-W	24	24	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE WELDS ABOVE GROUND SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 CODE: ASME B31.8 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 100 °C	12/16/2024	BY OZAC	U04	Page 2 of 4	150K5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
1200			RED ECC	9004136	20	20	REDUCER ECC WT 9.53 MM / 8.74 LTCS A860 WPHY-60-W	24	24		
1210											
1210	U04		OLETS	5504001	0.75	1.5	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2 CL.1	2	2		
1220	U04		OLETS	5504002	0.75	2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2 CL.1	3	3		
1230			OLETS	9006800	0.75	2	NIPOFLANGE 150# RF 6.02 MM/XS WT LTCS A694 F60	4	4		
1240			OLETS	9006801	0.75	2	NIPOFLANGE 150# RF 6.35 MM/XS WT LTCS A694 F60	6	14		
1250			OLETS	9006804	0.75	2	NIPOFLANGE 150# RF 7.92 MM/XS WT LTCS A694 F60	16	18		
1260			OLETS	9006805	0.75	2	NIPOFLANGE 150# RF 8.74 MM/XS WT LTCS A694 F60	20	20		
1270			OLETS	9006806	0.75	2	NIPOFLANGE 150# RF 9.53 MM/XS WT LTCS A694 F60	24	24		
1280			OLETS	9002639	0.75	2	NIPOFLANGE 300# RF 6.02 MM/XS WT LTCS A694 F60	4	4		
1290			OLETS	9002643	0.75	2	NIPOFLANGE 300# RF 6.35 MM/XS WT LTCS A694 F60	6	14		
1300			OLETS	9002645	0.75	2	NIPOFLANGE 300# RF 7.92 MM/XS WT LTCS A694 F60	16	18		
1310			OLETS	9002646	0.75	2	NIPOFLANGE 300# RF 8.74 MM/XS WT LTCS A694 F60	20	20		
1320			OLETS	9002647	0.75	2	NIPOFLANGE 300# RF 9.53 MM/XS WT LTCS A694 F60	24	24		
1330											
1330	U04		OLETS	5505335	0.75	1	WELDOLET SCH XS LTCS A350 LF2 CL.1	2	2		
1340	U04		OLETS	5505179	0.75	2	WELDOLET SCH STD/XS LTCS A350 LF2 CL.1	3	3		
1350	U04		OLETS	9006219	0.75	1.5	WELDOLET WT 6.02MM/SCH XS LTCS A694 F60	4	4		
1360			OLETS	9006407	0.75	2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6	14		
1370			OLETS	9006409	3	3	WELDOLET WT 6.35 MM/STD WT LTCS A694 F60	8	14		
1380			OLETS	9006408	4	4	WELDOLET WT 6.35 MM/6.02 MM LTCS A694 F60	12	14		
1390			OLETS	9002552	0.75	2	WELDOLET WT 7.92MM/ XS WT LTCS A694 F60	16	18		
1400			OLETS	9006415	3	3	WELDOLET WT 7.92 MM/STD WT LTCS A694 F60	16	18		
1410			OLETS	9006414	4	4	WELDOLET WT 7.92 MM/6.02 MM LTCS A694 F60	16	18		
1420			OLETS	9004124	6	6	WELDOLET WT 7.92 MM/6.35 MM LTCS A694 F60	18	18		
1430			OLETS	9006423	0.75	2	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	20	20		
1440			OLETS	9006422	3	3	WELDOLET WT 8.74 MM/STD WT LTCS A694 F60	20	20		
1450			OLETS	9006421	4	4	WELDOLET WT 8.74 MM/6.02 MM LTCS A694 F60	20	20		
1460			OLETS	9002008	6	6	WELDOLET WT 8.74 MM/6.35 MM LTCS A694 F60	20	20		
1470			OLETS	9006429	0.75	2	WELDOLET WT 9.53 MM/XS WT LTCS A694 F60	24	24		
1480			OLETS	9006428	3	3	WELDOLET WT 9.53 MM/STD WT LTCS A694 F60	24	24		
1490			OLETS	9006427	4	4	WELDOLET WT 9.53 MM/6.02 MM LTCS A694 F60	24	24		
1500			OLETS	9006424	6	8	WELDOLET WT 9.53 MM/6.35 MM LTCS A694 F60	24	24		
1510	U04			9008202	1	1	WELDOLET FLAT BOTTOM SCH XS LTCS A694 F60	0	0		
1520											
1520	U04	FA8	FLANGE	5599692	0.5	0.5	FLG SCRDR (FNPT) REDUCING 150# RF LTCS A350 LF2 CL.1	0.75	1.5		
1530	U04	FA8	TEE RED	5599703	0.5	0.5	FLG SCRDR (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1	0.75	0.75		
1540											
1540	U04		FLG ADAPTER	5851200	0.5	2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE	0	0		
1550	U04		FLG ADAPTER	5851203	0.75	0.75	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE	0	0		
1560											
1560	U04		FLANGE	5556264	0.5	2	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH XS	0	0		
1570	U04		FLANGE	5556261	3	3	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH STD	0	0		
1580			FLANGE	9006702	4	4	FLG WN 150# RF LTCS A694 F60 WT 6.02MM	0	0		
1590	U04	817	FLANGE	5556290	2	2	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH SCH 160	0	0		
1600			FLANGE	9001473	6	14	FLG WN 150# RF LTCS A694 F60 WT 6.35MM	0	0		
1610	U04		FLANGE	5556476	0.5	2	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0		
1620	U04		FLANGE	5556473	3	3	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0		
1630			FLANGE	9001480	4	4	FLG WN 300# RF LTCS A694 F60 WT 6.02MM	0	0		
1640	U04		FLANGE	9001483	6	14	FLG WN 300# RF LTCS A694 F60 WT 6.35MM	0	0		
1650	U04		FLANGE	5557036	0.75	2	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0		
1660	U04		FLANGE	5557033	3	3	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD	0	0		
1670			FLANGE	9001723	4	4	FLG WN 600# RF LTCS A694 F60 WT 6.02MM	0	0		
1680	U04	FS1, 817	FLANGE	5558036	2	2	FLG WN 600# RTJ LTCS A350 LF2 CL.1 SCH XS	0	0		
1690	U04	FS1	FLANGE	5555935	3	3	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH STD	0	0		
1700			FLANGE	9005389	4	4	FLG WN 900# RF LTCS A694 F60 WT 6.02MM	0	0		
1710	U04	FS1	FLANGE	5557393	1.5	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XS	0	0		
1720	U04	FS1, 817	FLANGE	5557507	2	2	FLG WN 1500# RTJ LTCS A350 LF2 CL.1 SCH XS	0	0		
1730	U04		BLIND	5615566	0.5	24	FLG BLIND 150# RF LTCS A350 LF2 CL.1	0	0		
1740	U04		BLIND	5615576	0.5	24	FLG BLIND 300# RF LTCS A350 LF2 CL.1	0	0		
1750	U04	FS1		5556282	0.75	0.75	FLG WN REDUCING 1500# RF A350 LF2 CL.1 BORED TO SUIT XS	1	1		
1760											
1760	U04		ORIFICE FLG	5623947	2	2	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0		
1770	U04		ORIFICE FLG	5623944	3	14	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0		
1780											
1780							***** GASKETS *****				
1790											
1790			GASKET	5673043	0.5	24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR	0	0		
1800			GASKET	5673033	0.5	24	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0		
1810			GASKET	5673103	0.75	20	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0		
1820	U04		GASKET	5673117	3	16	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE WELDS ABOVE GROUND SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 CODE: ASME B31.8 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 100 °C	12/16/2024	BY OZAC	U04	Page 3 of 4	150K5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1830			GASKET	5672251	1.5	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1840	U04	817	GASKET	5673202	2	2	GASKET 600# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING	0	0	
1850	U04	817	GASKET	5674491	3	3	GASKET 900# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING	0	0	
1860	U04	817	GASKET	5674492	2	2	GASKET 1500# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING	0	0	
1870										
1870							***** BOLTING *****			
1880										
1880	U04	B1	BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1890										
1890							**** BLINDS & SPACERS ****			
1900										
1900			SPEC BLIND	5749580	0.75	14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70	0	0	
1910			SPEC BLIND	5749581	1.5	10	SPECTACLE BLIND 300# RF LTCS A516 GR.70	0	0	
1920										
1920			PADDLE	5750023	16	24	PADDLE BLANK 150# RF LTCS A516 GR.70	0	0	
1930			PADDLE	5750100	12	24	PADDLE BLANK 300# RF LTCS A516 GR.70	0	0	
1940			PADDLE	5750240	16	24	PADDLE SPACER 150# RF LTCS A516 GR.70	0	0	
1950			PADDLE	5750200	12	24	PADDLE SPACER 300# RF LTCS A516 GR.70	0	0	
1960										
1960	U04		FLANGE	9006750	3	3	FLG WN REDUCING 150# RF LTCS A694 F60 BORED TO SUIT WT 6.02MM	4	4	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE WELDS ABOVE GROUND SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 CODE: ASME B31.8 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 100 °C	12/16/2024	BY OZAC	U04	Page 4 of 4	150K5F